SELECTED WORK · 2026

Thanapat Aupprathumwipanon

Software Engineer

Computer Engineering student at Chulalongkorn University. I work on things that go into production — a mentoring platform used by **1,000+ people**, registration sites for **5,000+ students**, and a reporting API for a **Thai government agency**.

Bangkok, Thailand · +66 99-009-0809 · thanapat.aupp@gmail.com

github.com/miramsalp · personal-website-miramsalp.vercel.app

01

Wise Mentoring Ship

Software Engineer Intern, Skooldio · Feb – Jul 2026 → Software Engineer (Part-time), Wise · Aug 2026 – now · 1,000+ mentors and mentees

An on-demand mentoring platform for Thai universities. A student books a session, a mentor accepts, and an admin can see the whole queue.

The five-step booking flow — one of the parts I built front end through to the API.

WHAT I BUILT

— Registration, mentee requests, mentor handling, the dashboard, the explore pages, and the admin back-office.

— Migration scripts that moved records already in use into the new schema.

— CI/CD, and LINE notifications sent through GCP Cloud Tasks so a slow push never blocks a user action.

— The design tokens and shared components the rest of the UI was built from.

AFTER THE CONTRACT

— Wise hired me directly when the Skooldio engagement ended. I still own the backend, and still ship to it.

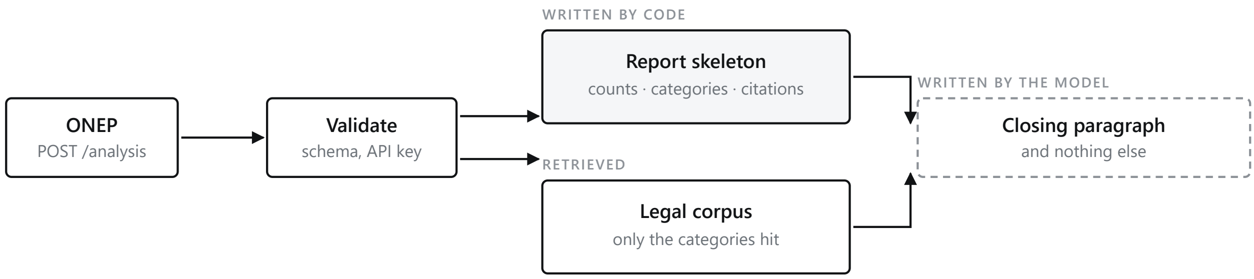
React · NestJS · TypeScript · PostgreSQL · GCP · LINE API

RAG-GIS Heritage Impact System

RAG-GIS System for Assessing Impacts on Art and Cultural Heritage Sites

Capstone project · Contributor — the `/analysis` endpoint, and SSE streaming for the chat API · Client: ONEP, a Thai government agency

Before a project breaks ground, ONEP checks whether it sits too close to protected natural sites and art and cultural heritage sites. That check produces numbers. This API turns them into the Thai report an officer would otherwise write by hand.



WHY IT IS BUILT THIS WAY

- Code writes every count and every legal citation. The model only writes the last paragraph, so no number in the report can be invented.
- ONEP retries a 5xx but gives up on a 4xx. So an empty answer from the model raises an error instead of returning half a report with a 200.
- The endpoint gives up at 100 seconds, inside ONEP's 120-second timeout. A hang becomes a retry, not a lost request.
- Unknown fields are ignored rather than rejected, so a newer payload from them still works.

WHAT I CONTRIBUTED

- The request schemas and the controller for the endpoint ONEP calls.
- The report assembly, and the retrieval query built from the site categories a project actually hits.
- SSE token streaming for the chat API, so an answer appears as it is written instead of all at once at the end.

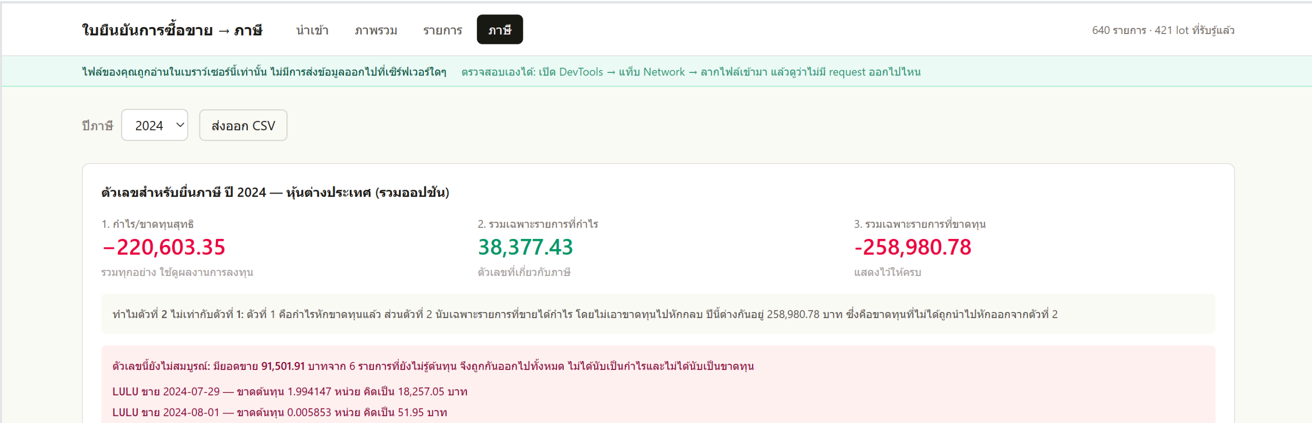
Python · FastAPI · Pydantic · Vector search · SSE

03

Tax Tracker

Solo · Built for my own tax filing

Thai investors have to report realised gains on foreign shares, and the broker only hands over PDFs — one per trade. Drop them into the page and it does the FIFO accounting.



Realised profit and loss by month, computed from the confirmation notes.

WHY IT IS BUILT THIS WAY

- Nothing is uploaded. Parsing runs in the browser and the CSP blocks outgoing requests — the banner at the top tells you how to check that yourself.
- The parser refuses a PDF it does not recognise. A wrong number here ends up on a tax return.
- It shows what it could not work out: the sells with unknown cost basis are listed in red, not quietly averaged into the total.
- Confirmation notes are checked against the monthly statement, so a missing trade shows up instead of shifting every lot after it.

TypeScript · Vite · In-browser PDF parsing · CSP

04

First Date & RubpuenKaomai 2026

Frontend Developer, ISD — Chulalongkorn University · Jul 2026

Two freshman-welcome sites for Chulalongkorn student organisations. Around 5,000 students sign up, most of them in the first hour.

- Registration flow, landing page and edit-profile screen.
- The API integration behind all three.
- Mobile-first: almost every sign-up arrives on a phone.

React

05

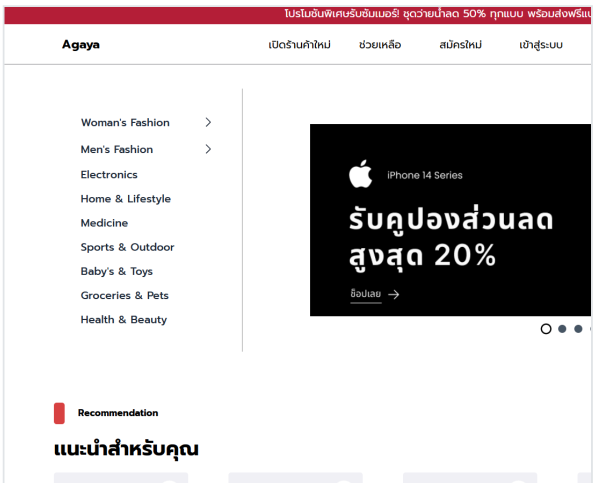
Agaya

Team Lead · group project

A marketplace sitting between vendors and buyers, handling the listing, the order and the money in between.

- Ran the sprints and owned delivery across the stack.
- Built the Stripe payment flow and Cloudinary image handling.

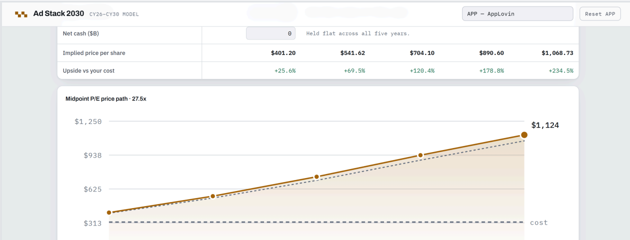
Stripe · Cloudinary



06

Also built

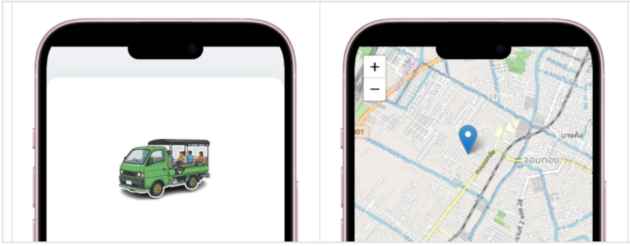
Side projects, each one built to answer a question I actually had



Ad Stack 2030

An equity model for 37 companies. Edit the growth and margin assumptions and every projection, scenario and IRR through 2030 re-runs in the browser.

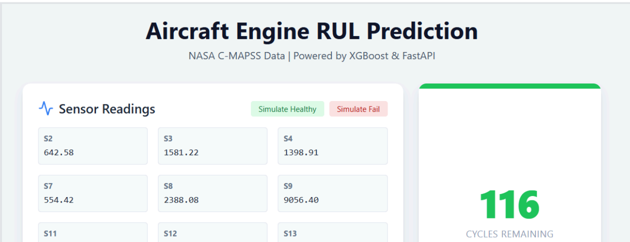
React 19 · Vite



Krapong-Go

Real-time transit for Thonburi, where I grew up. Passengers signal that they are waiting and drivers see it live, instead of guessing and running empty.

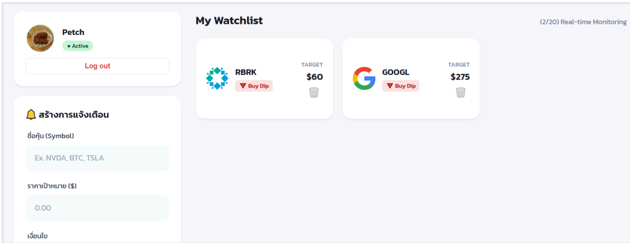
Socket.IO · Redis · MongoDB · Docker



Predictive Maintenance

Predicts how many cycles a turbofan engine has left, from NASA sensor data. XGBoost, RMSE 16.78 and R^2 0.82, served through a live dashboard.

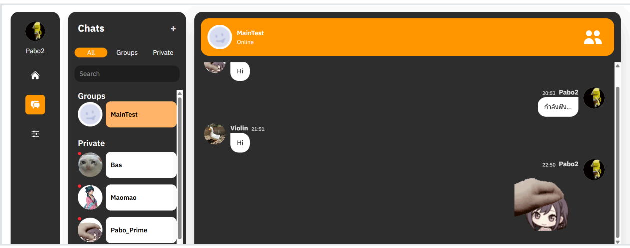
Python · XGBoost · FastAPI · React



Stock Watcher

Price alerts pushed into a LINE group. Used daily by friends and family who were otherwise checking prices by hand.

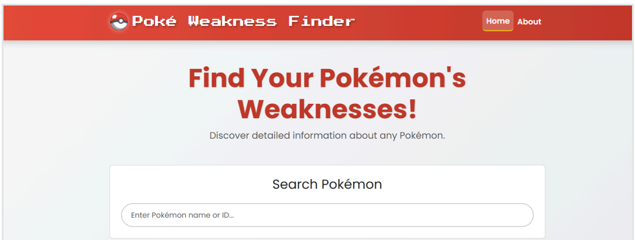
LINE Messaging API · LINE Login



Otto Webchat

A group chat app, built mostly to learn the deployment half: a load balancer on AWS Elastic Beanstalk and a custom domain in front of it.

AWS · Elastic Beanstalk · Node.js



Smaller experiments

A retriever over Thai law, chunked by legal article and scored with Hit@k and Recall@k — the groundwork behind 02. Plus an FPGA VGA switcher and a Pokémon weakness finder.

Python · BM25 · Verilog